In [2]:
```
import pandas as pd
import numpy as np
a = [10,20,30,40,50]
b = [40,50,60,70,80]
c = pd.Series(a,b)
un = pd.Series(np.intersect1d(a,b))
un
```

Out[2]:
```
0    40
1    50
dtype: int32
```

In [3]:
```
data = {"ename":['roshan','amar','ashwini','lohith','mohan','promodh'],
        'type':['regular','adhoc','regular','adhoc','contract','regular'],
        'Dname':['cs','cs','ec','ec','cs','ec'],
        'exp':[10,20,5,14,9,8],
        'salary':[50000,15000,30000,15000,10000,40000]}
df = pd.DataFrame(data)
df
```

Out[3]:

	ename	type	Dname	exp	salary
0	roshan	regular	cs	10	50000
1	amar	adhoc	cs	20	15000
2	ashwini	regular	ec	5	30000
3	lohith	adhoc	ec	14	15000
4	mohan	contract	cs	9	10000
5	promodh	regular	ec	8	40000

In [4]:
```
table = pd.pivot_table(df,index=['Dname'],values='salary',aggfunc=['mean'])
table
```

Out[4]:

	mean
	salary
Dname	
cs	25000.000000
ec	28333.333333

In [5]:
```
pivot = pd.pivot_table(
    data=df,
    index='type',
    values='salary',
    aggfunc=['mean', 'sum']
)

print(pivot)
```

```
           mean     sum
         salary  salary
type
adhoc     15000   30000
contract  10000   10000
regular   40000  120000
```

In [6]:
```
df.pivot_table(index='type',values='salary',aggfunc=len)
```

Out[6]:

type	salary
adhoc	2
contract	1
regular	3

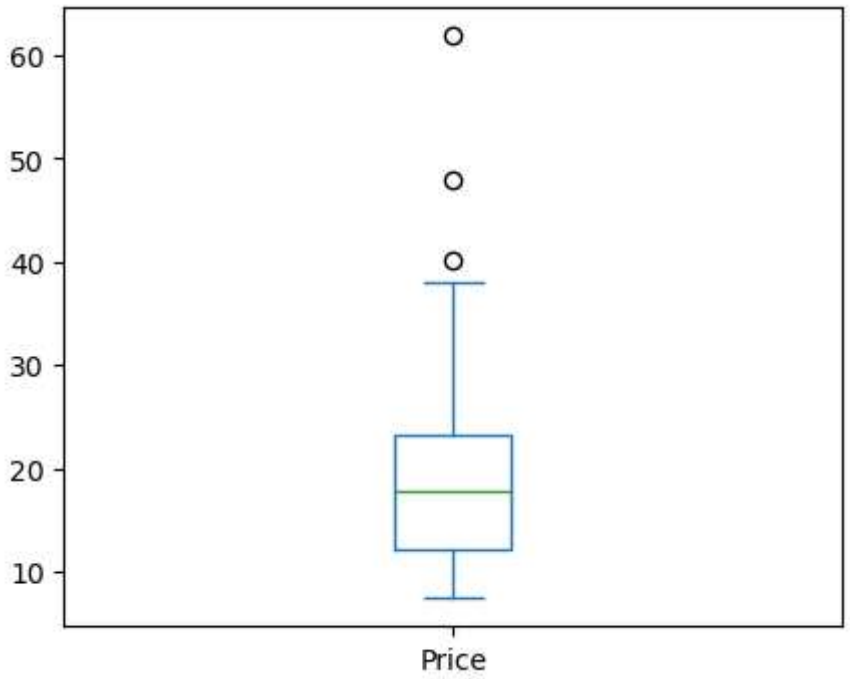
In [7]:
```
cars_df = pd.read_csv("Cars93.csv")
columns = ["Manufacturer","Model","Type","Price","MPG.city","MPG.highway","Horsepo
cars_df[columns].head()
```

Out[7]:

	Manufacturer	Model	Type	Price	MPG.city	MPG.highway	Horsepower	Rear.seat.room
0	Acura	Integra	Small	15.9	25	31	140	26.5
1	Acura	Legend	Midsize	33.9	18	25	200	30.0
2	Audi	90	Compact	29.1	20	26	172	28.0
3	Audi	100	Midsize	37.7	19	26	172	31.0
4	BMW	535i	Midsize	30.0	22	30	208	27.0

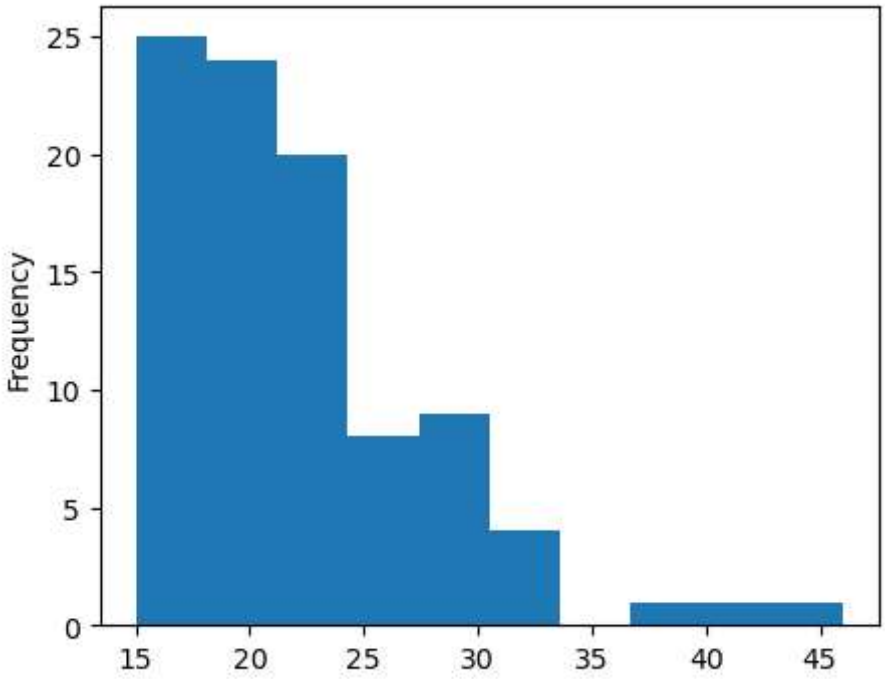
In [8]:
```
cars_df["Price"].plot(kind="box",figsize = (5,4))
```

Out[8]:
```
<AxesSubplot:>
```



In [10]:
```
cars_df["MPG.city"].plot(kind="hist",figsize = (5,4))
```

Out[10]:
```
<AxesSubplot:ylabel='Frequency'>
```

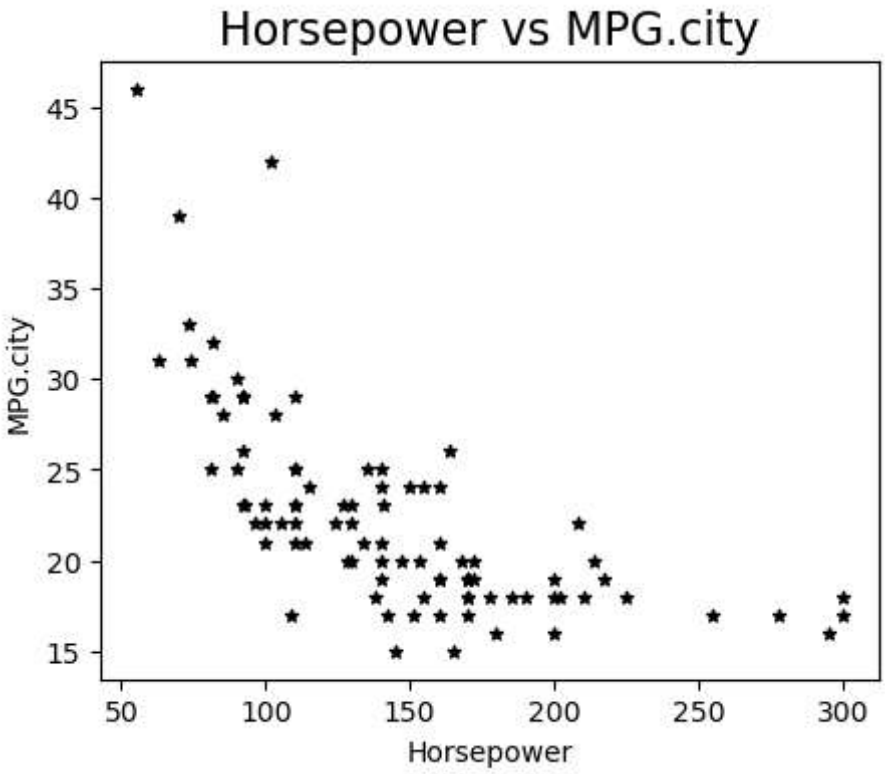


In [45]:
```
ax = cars_df.plot(["Horsepower"],["MPG.city"],kind="scatter", color = "black",m

#To add labels and title to the output

ax.set_xlabel("Horsepower")     #sets label for x-axis
ax.set_ylabel("MPG.city")       #sets label for y-axis
ax.set_title("Horsepower vs MPG.city",fontsize=16)      #sets title for the graph
```

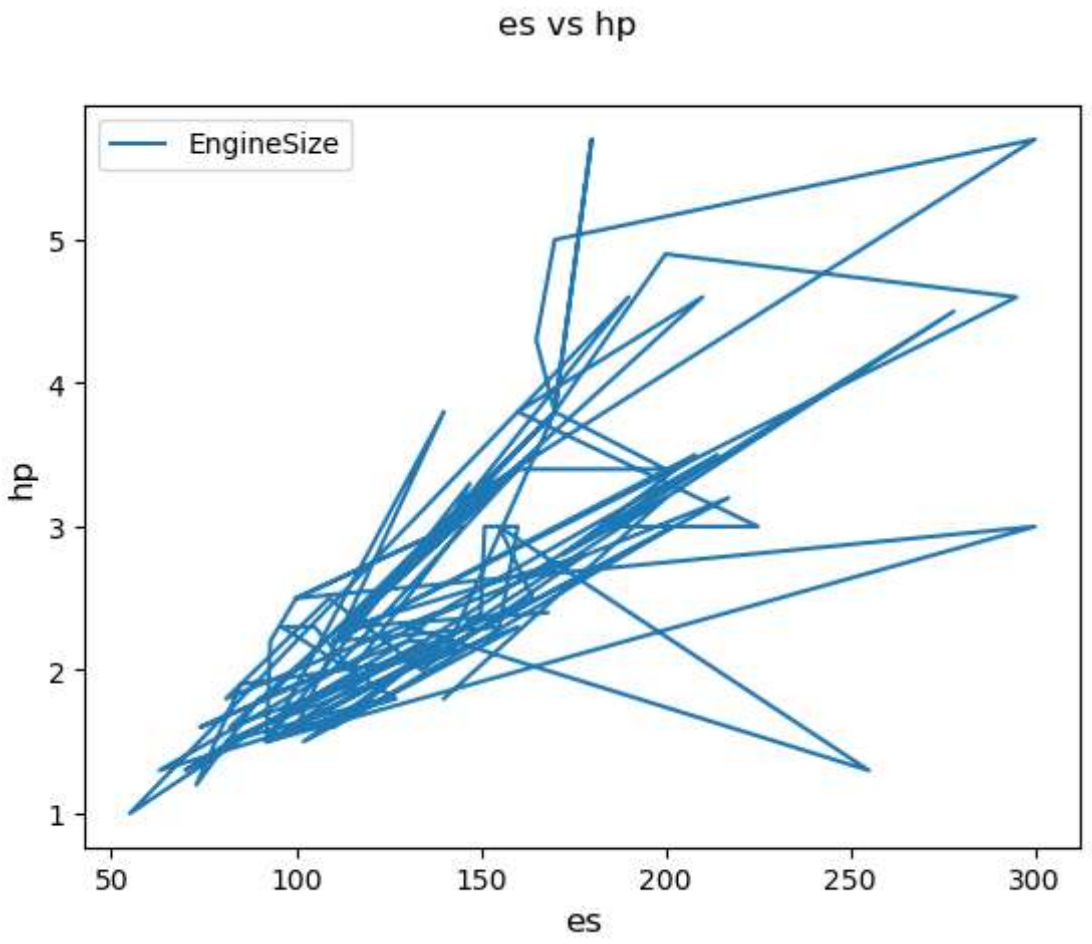
Out[45]:
```
Text(0.5, 1.0, 'Horsepower vs MPG.city')
```



In [67]:
```
import matplotlib.pyplot as plt
plt.figure(figsize=(20,20))
cars_df.plot(x='Horsepower', y='EngineSize')
plt.suptitle('es vs hp', fontsize=12, color='black')
plt.xlabel("es",fontsize=12, color='black')
```

```
plt.ylabel("hp", fontsize=12, color='black')
plt.show()
```

```
<Figure size 2000x2000 with 0 Axes>
```



In [60]:
```
my_list= [1,3,5,7,9,11]
my_list_squared = list(map(lambda i: i**2, my_list))
my_list_squared
```

Out[60]:
```
[1, 9, 25, 49, 81, 121]
```

In [52]:
```
list1 = [4,5,6,7,8,9,13,55]
oddlist = list(filter(lambda x:False if x%2==0 else True,list1))
oddlist
```

Out[52]:
```
[5, 7, 9, 13, 55]
```

In [61]:
```
from functools import reduce
def sum(x,y):
    return x+y

list1 = [1,2,3,4,5,56]
s = reduce(sum,list1)
s
```

Out[61]:
```
71
```

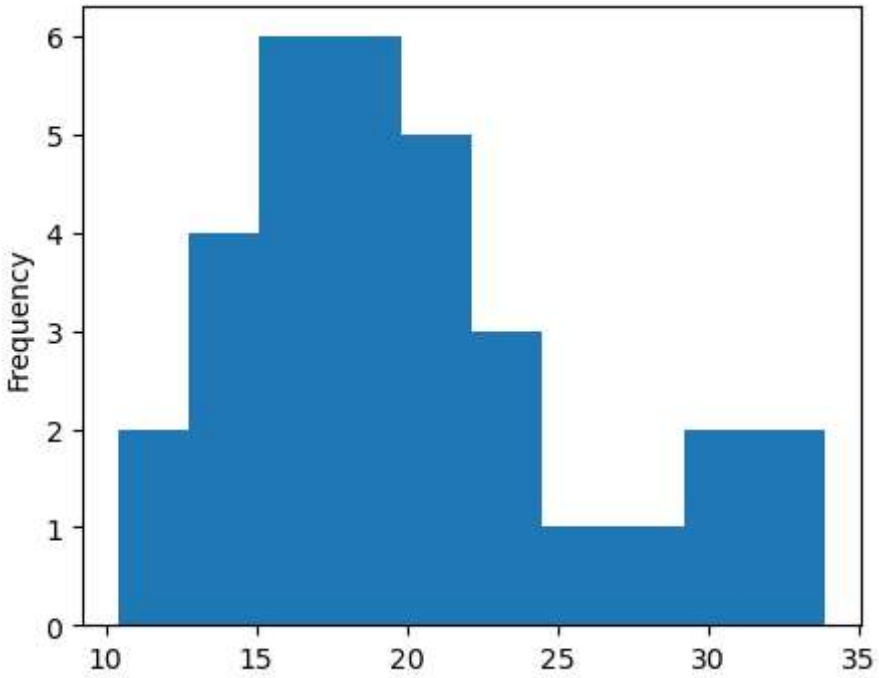
In [11]:
```
mcars_df = pd.read_csv("mtcars.csv")
columns = ["model","mpg","cyl","disp","hp","drat","wt","qsec","vs","am","gear","ca
mcars_df[columns].head(10)
```

Out[11]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
8	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
9	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

In [12]:
```
mcars_df["mpg"].plot(kind="hist",figsize = (5,4))
```

Out[12]:
```
<AxesSubplot:ylabel='Frequency'>
```

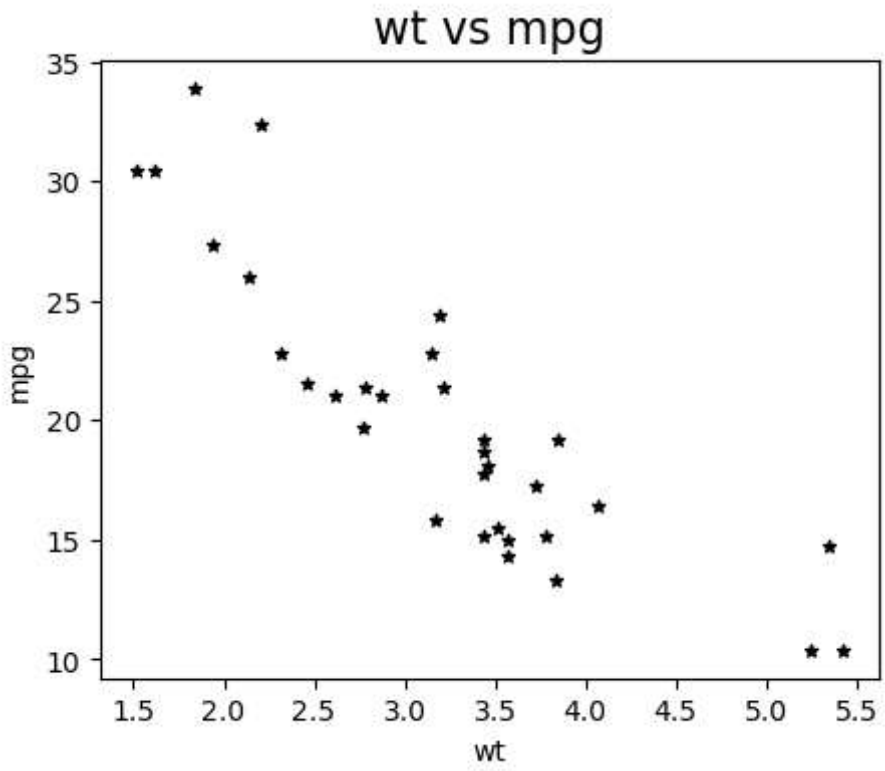


In [57]:
```
ax1 = mcars_df.plot(["wt"],["mpg"],kind="scatter", color = "black",marker = "*"

#To add labels and title to the output

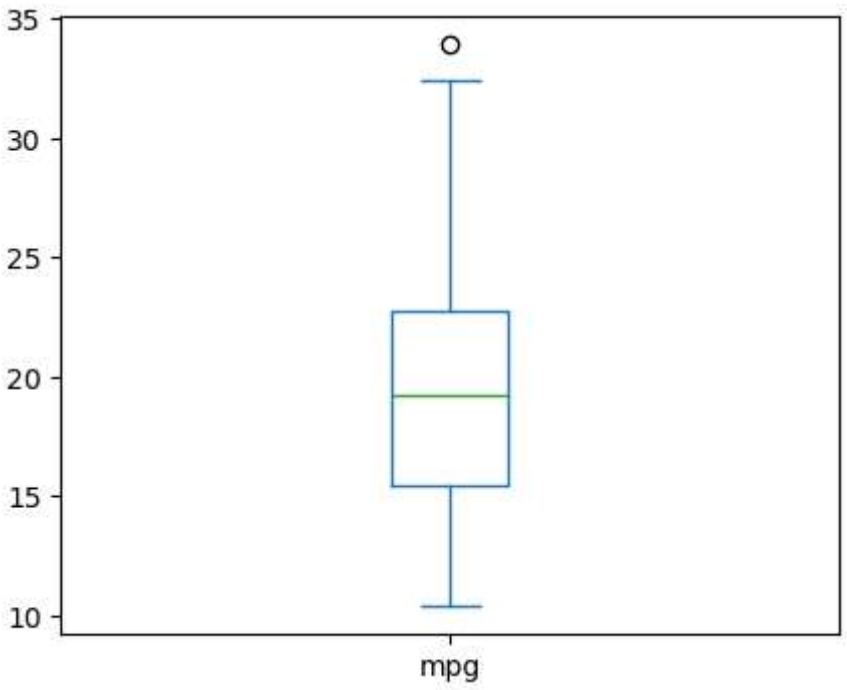
ax1.set_xlabel("wt")    #sets label for x-axis
ax1.set_ylabel("mpg")   #sets label for y-axis
ax1.set_title("wt vs mpg",fontsize=16)  #sets title for the graph
```

Out[57]:
```
Text(0.5, 1.0, 'wt vs mpg')
```



In [58]:
```
mcars_df["mpg"].plot(kind="box",figsize = (5,4))
```

Out[58]:
```
<AxesSubplot:>
```



In [68]:
```
data = {
  'book_name': ['book1','book2','book3','book4','book1','book2','book3','book5'],
  'book_type': ['math','physics','computer','science','math','physics','computer',
  'book_id': [1,2,3,4,1,2,3,5]
}

df = pd.DataFrame(data)
df

print(df.groupby(["book_name"]).count())
```

book_name	book_type	book_id
book1	2	2
book2	2	2
book3	2	2
book4	1	1
book5	1	1

In [ ]: